

Question 1:

Find the cofactor matrix for

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

Use the cofactor matrix to find A^{-1} .

$$C = \begin{bmatrix} \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} & -\begin{vmatrix} -1 & -1 \\ 0 & 2 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 0 & -1 \end{vmatrix} \\ -\begin{vmatrix} -1 & 0 \\ -1 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 0 & -1 \end{vmatrix} \\ \begin{vmatrix} -1 & 0 \\ 2 & -1 \end{vmatrix} & -\begin{vmatrix} 2 & 0 \\ -1 & -1 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\therefore \text{adj}(A) = C^T = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\therefore \det(A) = 0 \begin{vmatrix} -1 & 0 \\ 2 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 2 & 0 \\ -1 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4$$

$$\therefore A^{-1} = \frac{\text{adj}(A)}{\det(A)} = \begin{bmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{bmatrix}$$

Question 2:

Find the determinant of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 6 & 6 & 1 \\ -1 & 0 & 0 & 3 \\ 0 & 2 & 0 & 5 \end{bmatrix}.$$

Find the determinants of A^{-1} and A^2 .

$$\det(A) = -1 \cdot \begin{vmatrix} 2 & 3 & 0 \\ 6 & 6 & 1 \\ 2 & 0 & 5 \end{vmatrix} - 0 + 0 - 3 \cdot \begin{vmatrix} 1 & 2 & 3 \\ 2 & 6 & 6 \\ 0 & 2 & 0 \end{vmatrix}$$

$$= -1 \cdot (2 \cdot \begin{vmatrix} 6 & 1 \\ 0 & 5 \end{vmatrix} - 3 \cdot \begin{vmatrix} 6 & 1 \\ 2 & 5 \end{vmatrix} + 0)$$

$$- 3 \cdot (0 - 2 \cdot \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} + 0)$$

$$= -1 \cdot (2 \cdot 30 - 3 \cdot 28) - 3 \cdot (-2 \cdot 0)$$

$$= 24$$

$$\det(A^{-1}) = \frac{\det(I)}{\det(A)} = \frac{1}{\det(A)} = \frac{1}{24}$$

$$\det(A^2) = \det(A) \det(A) = 24^2 = 576$$

Question 3:

Find the nullspace of the following matrices

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 1 & 2 \\ 6 & -1 & 1 \\ 3 & 2 & 5 \end{bmatrix}.$$

$$(1) \quad Ax = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ 1 \\ x_3 \end{bmatrix} = 0$$

$$\therefore S_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

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$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 1 & 2 \\ 6 & -1 & 1 \\ 3 & 2 & 5 \end{bmatrix}.$$

$\therefore$ The nullspace is $C(B)$

$$\text{where } B = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

The nullity of A is 1

Question 3:

Find the nullspace of the following matrices

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 1 & 2 \\ 6 & -1 & 1 \\ 3 & 2 & 5 \end{bmatrix}.$$

(2)

$$Ax = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = 0$$

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Find the nullspace of the following matrices

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 1 & 2 \\ 6 & -1 & 1 \\ 3 & 2 & 5 \end{bmatrix}.$$

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ 1 \\ x_3 \\ 0 \\ 0 \end{bmatrix} = 0 \quad \therefore S_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 1 \\ 0 \end{bmatrix} = 0 \quad \therefore S_2 = \begin{bmatrix} 0 \\ -2 \\ 1 \\ -2 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ 0 \\ x_3 \\ 0 \\ 1 \end{bmatrix} = 0 \quad \therefore S_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -3 \\ 1 \end{bmatrix}$$

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Find the nullspace of the following matrices

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$\therefore$ The nullspace is $C(B)$

$$\text{where } B = \begin{bmatrix} -2 & -2 & 0 \\ 1 & 1 & 1 \\ 0 & -2 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The nullity of A is 3

Question 3:

Find the nullspace of the following matrices

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$$(3) \quad Ax = \begin{bmatrix} 3 & 1 & 2 \\ 6 & -1 & 1 \\ 3 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 3 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} 3 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore$ The nullspace is $C(B)$

$$\text{where } B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The nullity of A is 0

Question 4:

There is a fundamental relationship between the null and column spaces of a matrix A and its transpose A^T :

- i) Show that if $x \in \mathcal{N}(A^T)$ and $y \in \mathcal{C}(A)$, then $x^T y = 0$.
- ii) Likewise, show that if $x \in \mathcal{N}(A)$ and $y \in \mathcal{C}(A^T)$, then $x^T y = 0$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \therefore \mathcal{C}(A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \quad \therefore \mathcal{C}(A^T) = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\text{i)} \quad \begin{bmatrix} a & c \\ b & d \end{bmatrix} B = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} b_1 & b_3 \\ b_2 & b_4 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} ab_1 + cb_3 & ab_3 + cb_4 \\ bb_1 + db_3 & bb_3 + db_4 \end{bmatrix} = 0$$

$$B^T = \begin{bmatrix} b_1 & b_3 \\ b_2 & b_4 \end{bmatrix}$$

$$\therefore B^T A = \begin{bmatrix} b_1 & b_3 \\ b_2 & b_4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ab_1 + cb_3 & ab_3 + cb_4 \\ bb_1 + db_3 & bb_3 + db_4 \end{bmatrix} = 0$$

Question 4:

There is a fundamental relationship between the null and column spaces of a matrix A and its transpose A^T :

i) Show that if $x \in \mathcal{N}(A^T)$ and $y \in \mathcal{C}(A)$, then $x^T y = 0$.

ii) Likewise, show that if $x \in \mathcal{N}(A)$ and $y \in \mathcal{C}(A^T)$, then $x^T y = 0$.

$$(ii) \begin{bmatrix} a & b \\ c & d \end{bmatrix} B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} ab_1 + bb_2 & ab_3 + bb_4 \\ cb_1 + db_2 & cb_3 + db_4 \end{bmatrix} = 0$$

$$B^T = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$$

$$\therefore B^T A^T = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} ab_1 + bb_2 & ab_3 + bb_4 \\ cb_1 + db_2 & cb_3 + db_4 \end{bmatrix} = 0$$

(i) If $y \in \mathcal{C}(A)$, there must be some z such that
 $y = Az$

$$\text{Then } x^T y = x^T Az = (xA^T)^T z = 0 \cdot z = 0$$

(ii) $y = A^T z$

$$\text{Then } x^T y = x^T A^T z = (xA)^T z = 0 \cdot z = 0$$

Question 5:

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 0 & 6 & 1 \\ 0 & 1 & 6 \end{bmatrix}.$$

$$Au = \lambda u = \lambda Iu$$

$$\therefore (A - \lambda I)u = 0 \quad \therefore \det(A - \lambda I) = 0$$

$$\therefore \det \begin{bmatrix} 3-\lambda & 0 & 4 \\ 0 & 6-\lambda & 1 \\ 0 & 1 & 6-\lambda \end{bmatrix} = (3-\lambda) \det \begin{bmatrix} 6-\lambda & 1 \\ 1 & 6-\lambda \end{bmatrix}$$

$$= (3-\lambda)(\lambda^2 - 12\lambda + 36 - 1)$$

$$= (3-\lambda)(\lambda^2 - 12\lambda + 35)$$

$$= -\lambda^2 + 15\lambda^2 - 71\lambda + 105$$

$$= (-\lambda + 7)(\lambda^2 - 8\lambda + 15)$$

$$= (-\lambda + 7)(\lambda - 3)(\lambda - 5)$$

$$= 0$$

Question 5:

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 0 & 6 & 1 \\ 0 & 1 & 6 \end{bmatrix}.$$

$$\therefore \lambda_1 = 7, \lambda_2 = 3, \lambda_3 = 5$$

$$\therefore (A - \lambda I)u = 0$$

For $\lambda_1 = 7$,

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} u_1 = \begin{bmatrix} -4 & 0 & 4 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} u_{1a} \\ u_{1b} \\ u_{1c} \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} -4 & 0 & 4 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_{1a} \\ u_{1b} \\ u_{1c} \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} -4 & 0 & 4 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_{1a} \\ u_{1b} \\ 1 \end{bmatrix} = 0 \quad \therefore u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 3$,

$$\begin{bmatrix} 0 & 0 & 4 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix} u_2 = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} u_{2a} \\ u_{2b} \\ u_{2c} \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} 0 & 0 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & \frac{8}{3} \end{bmatrix} \begin{bmatrix} u_{2a} \\ u_{2b} \\ u_{2c} \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} 0 & 0 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & \frac{8}{3} \end{bmatrix} \begin{bmatrix} 1 \\ u_{2b} \\ u_{2c} \end{bmatrix} = 0 \quad \therefore u_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

Question 5:

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 0 & 6 & 1 \\ 0 & 1 & 6 \end{bmatrix}.$$

For $\lambda_3 = 5$,

$$\begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} u_3 = \begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_{3a} \\ u_{3b} \\ u_{3c} \end{bmatrix} = 0 \quad \therefore \begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_{3a} \\ u_{3b} \\ u_{3c} \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} -2 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} u_{3a} \\ u_{3b} \\ 1 \end{bmatrix} = 0 \quad \therefore u_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore \lambda_1 = 7, \lambda_2 = 3, \lambda_3 = 5$$

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Question 6:

Suppose that the eigenvalues and eigenvectors of a matrix A are known. What are the eigenvalues and eigenvectors of $A + I$?

Verify your answer by computing the eigenvalues and eigenvectors of A and $A + I$, where

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}.$$

$$\lambda_{A+I} = \lambda_A + 1$$

$$u_{A+I} = u_A$$

$$\therefore Au = \lambda u = \lambda I u \quad \therefore (A - \lambda I)u = 0$$

$$\therefore \det(A - \lambda I) = 0 \quad \therefore A - \lambda I = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

$$\therefore \det(A - \lambda I) = \lambda^2 - 4\lambda + 3 - 8 = (\lambda - 5)(\lambda + 1) = 0$$

$$\therefore \lambda_1 = 5, \lambda_2 = -1$$

$$\text{For } \lambda_1 = 5, \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} u_1 = 0 \quad \therefore u_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_2 = -1, \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} u_2 = 0 \quad \therefore u_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Question 6:

Suppose that the eigenvalues and eigenvectors of a matrix A are known. What are the eigenvalues and eigenvectors of $A + I$?

Verify your answer by computing the eigenvalues and eigenvectors of A and $A + I$, where

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}.$$

Consider $A + I$

$$\therefore \det(A + I - \lambda I) = 0 \quad \because A + I - \lambda I = \begin{bmatrix} 2-\lambda & 4 \\ 2 & 4-\lambda \end{bmatrix}$$

$$\therefore \det(A - \lambda I) = \lambda^2 - 6\lambda + 8 - 8 = \lambda(\lambda - 6) = 0$$

$$\therefore \lambda_1 = 0, \lambda_2 = 6$$

$$\text{For } \lambda_1 = 0, \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} u_1 = 0 \quad \therefore u_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_2 = 6, \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} u_2 = 0 \quad \therefore u_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Question 7:

Suppose that two matrices A and B have eigenvalues α and β respectively. A common mistake is to assume that the eigenvalues of AB will be $\alpha\beta$. A false proof is

$$ABv = A(\beta v)$$

$$ABv = \beta(Av)$$

$$ABv = \beta\alpha v.$$

Explain why this proof is wrong. Under what circumstances will the claim be true?

Only when $U_A = U_B = v$, the claim be true

Question 8:

Find e^A , where

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

$$\begin{aligned} \det(A - I\lambda) &= \det \begin{bmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{bmatrix} = \lambda^2 - 2\lambda + 1 - 4 = \lambda^2 - 2\lambda - 3 \\ &= (\lambda - 3)(\lambda + 1) = 0 \end{aligned}$$

$$\therefore \lambda_1 = 3, \lambda_2 = -1 \quad \therefore \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$$

For $\lambda_1 = 3$,

$$(A - I\lambda)U_1 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} U_1 = 0 \quad \therefore U_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = -1$,

$$(A - I\lambda)U_2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} U_2 = 0 \quad \therefore U_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Question 8:

Find e^A , where

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

$$\therefore U^{-1} = \frac{\text{adj}(U)}{\det(U)} = \frac{\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}^T}{1 - (-1)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\therefore e^A = U e^{\Lambda} U^{-1}$$

$$= \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^3 & 0 \\ 0 & e^{-1} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} e^3 & -e^{-1} \\ e^3 & e^{-1} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}(e^3 + e^{-1}) & \frac{1}{2}(e^3 - e^{-1}) \\ \frac{1}{2}(e^3 - e^{-1}) & \frac{1}{2}(e^3 + e^{-1}) \end{bmatrix}$$

Question 9:

The Fibonacci numbers f_n satisfy the relationship

$$f_n = f_{n-1} + f_{n-2} \quad \text{for all } n = 2, \dots, \infty,$$

with $f_0 = 0$ and $f_1 = 1$. The first few Fibonacci numbers are 0, 1, 1, 2, 3, 5, 8, 13, 21, Find a general expression for f_n , and compute f_{100} .

Hint: Find a matrix A such that the relationship between consecutive pairs of Fibonacci numbers is

$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = A \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}.$$

$$\therefore A \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix} = \begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = \begin{bmatrix} f_n + f_{n-1} \\ f_n \end{bmatrix} \quad \therefore A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det(A - I\lambda) = \det \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 - \lambda - 1 = 0$$

$$\therefore \lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2} \quad \therefore A = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix}$$

$$\text{For } \lambda_1 = \frac{1+\sqrt{5}}{2},$$

$$\begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 1 & -\frac{1-\sqrt{5}}{2} \end{bmatrix} u_1 = 0 \quad \therefore u_1 = \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

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$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = A \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}.$$

$$\text{For } \lambda_2 = \frac{1-\sqrt{5}}{2},$$

$$\begin{bmatrix} \frac{1+\sqrt{5}}{2} & 1 \\ 1 & \frac{1-\sqrt{5}}{2} \end{bmatrix} u_2 = 0 \quad \therefore u_2 = \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned} \therefore U^{-1} &= \frac{\text{adj}(U)}{\det(U)} = \frac{\begin{bmatrix} 1 & -1 \\ -(\frac{1-\sqrt{5}}{2}) & \frac{1+\sqrt{5}}{2} \end{bmatrix}^T}{\frac{1+\sqrt{5}}{2} - (\frac{1-\sqrt{5}}{2})} \\ &= \frac{\begin{bmatrix} 1 & -(\frac{1-\sqrt{5}}{2}) \\ -1 & \frac{1+\sqrt{5}}{2} \end{bmatrix}}{\sqrt{5}} = \begin{bmatrix} \frac{\sqrt{5}}{5} & \frac{5-\sqrt{5}}{10} \\ -\frac{\sqrt{5}}{5} & \frac{5+\sqrt{5}}{10} \end{bmatrix} \end{aligned}$$

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Hint: Find a matrix A such that the relationship between consecutive pairs of Fibonacci numbers is

$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = A \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}.$$

$$\begin{aligned} \therefore \begin{bmatrix} f_{100} \\ f_{99} \end{bmatrix} &= A^{99} \begin{bmatrix} f_1 \\ f_0 \end{bmatrix} = U \Lambda^{99} U^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1+\sqrt{5}}{2} & \frac{1-\sqrt{5}}{2} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^{99} & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^{99} \end{bmatrix} \frac{\begin{bmatrix} 1 & -\left(\frac{1-\sqrt{5}}{2}\right) \\ -1 & \frac{1+\sqrt{5}}{2} \end{bmatrix}}{\sqrt{5}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^{100} & \left(\frac{1-\sqrt{5}}{2}\right)^{100} \\ \left(\frac{1+\sqrt{5}}{2}\right)^{99} & \left(\frac{1-\sqrt{5}}{2}\right)^{99} \end{bmatrix} \frac{\begin{bmatrix} 1 & -\left(\frac{1-\sqrt{5}}{2}\right) \\ -1 & \frac{1+\sqrt{5}}{2} \end{bmatrix}}{\sqrt{5}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^{100} - \left(\frac{1-\sqrt{5}}{2}\right)^{100} & \left(\frac{1+\sqrt{5}}{2}\right)^{99} - \left(\frac{1-\sqrt{5}}{2}\right)^{99} \\ \left(\frac{1+\sqrt{5}}{2}\right)^{99} - \left(\frac{1-\sqrt{5}}{2}\right)^{99} & \left(\frac{1+\sqrt{5}}{2}\right)^{98} - \left(\frac{1-\sqrt{5}}{2}\right)^{98} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^{100} - \left(\frac{1-\sqrt{5}}{2}\right)^{100} \\ \left(\frac{1+\sqrt{5}}{2}\right)^{99} - \left(\frac{1-\sqrt{5}}{2}\right)^{99} \end{bmatrix} \end{aligned}$$

Question 9:

The Fibonacci numbers f_n satisfy the relationship

$$f_n = f_{n-1} + f_{n-2} \quad \text{for all } n = 2, \dots, \infty,$$

with $f_0 = 0$ and $f_1 = 1$. The first few Fibonacci numbers are 0, 1, 1, 2, 3, 5, 8, 13, 21, Find a general expression for f_n , and compute f_{100} .

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$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = A \begin{bmatrix} f_n \\ f_{n-1} \end{bmatrix}.$$

$$\therefore f_{100} = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{100} - \left(\frac{1-\sqrt{5}}{2} \right)^{100} \right]$$